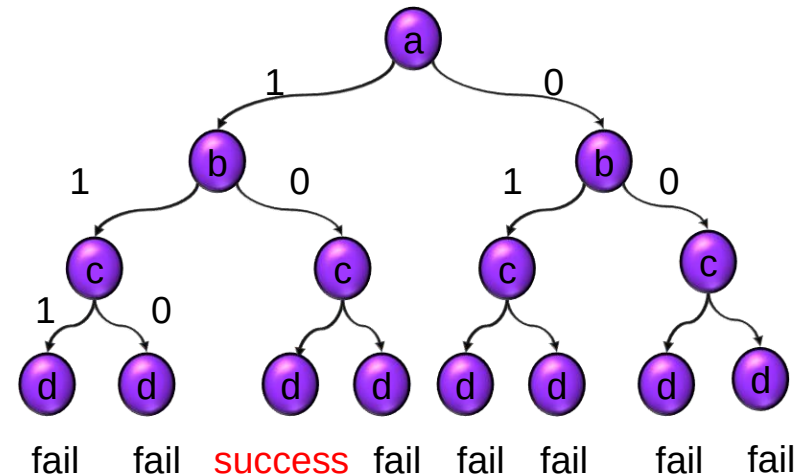


Combinational ATPG

- Introduction
- Deterministic Test Pattern Generation
 - ◆ Boolean difference *
 - ◆ Path sensitization **
 - ◆ D-Algorithm [Roth 1966]**
 - ◆ PODEM [Goel 1981]**
 - ◆ FAN [Fujwara 1985]**
 - ◆ SAT-based [Larrabee 92] *
- Acceleration techniques
- Concluding Remarks

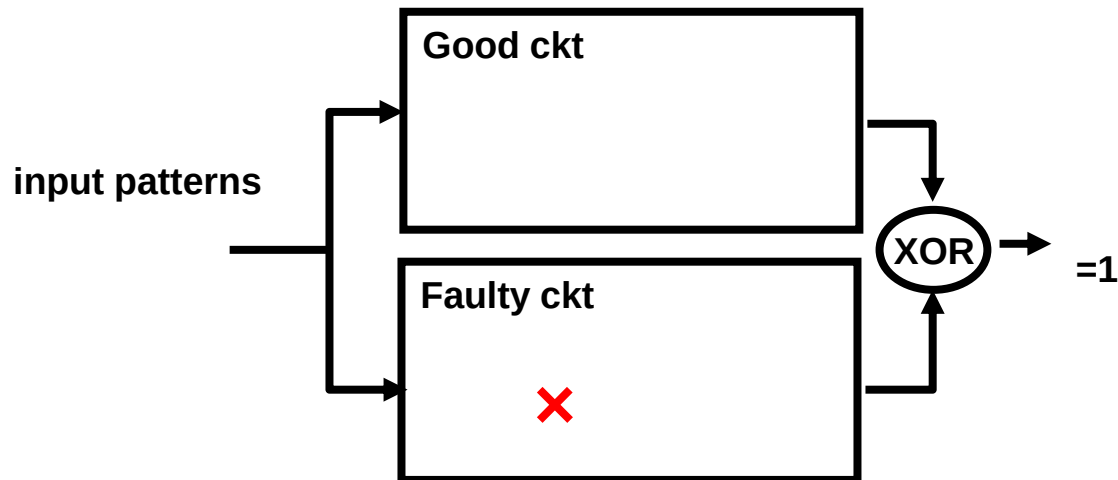
*Boolean-based methods

**path-based methods



SAT-based ATPG

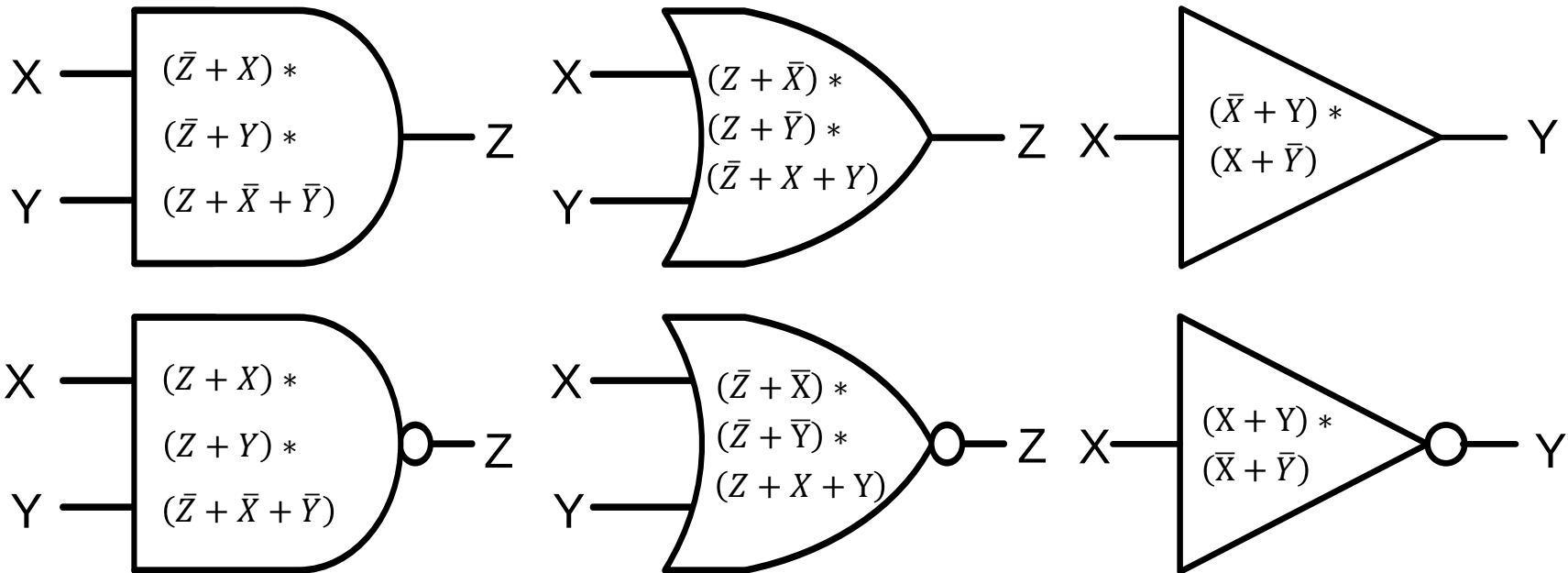
- Idea: convert test generation problem into *satisfiability problem*
 - ♦ Solve for a set of inputs such that
 - ♦ (good output) XOR (faulty output) = 1



this is called a *miter*

Tseitin Transformation [Tseitin 66]

- Converts circuit into **Conjunctive Normal Form (CNF)**
 - linear time, linear number of clauses and literals

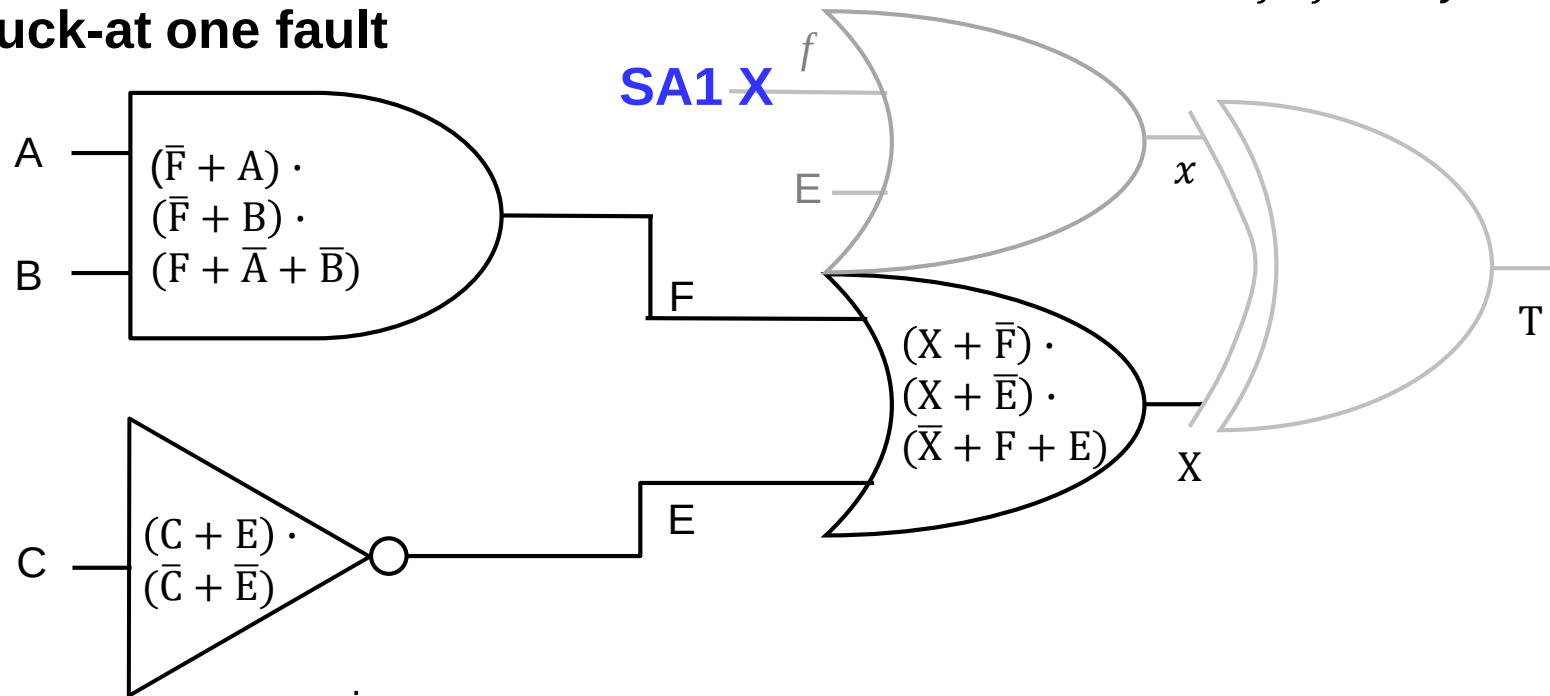


CNF for basic gates

Example (cont'd)

F good
 f faulty

- F stuck-at one fault



faulty

$$(x + \bar{f}) \cdot (x + \bar{E}) \cdot (\bar{x} + f + E) \cdot (f)$$

good

$$\begin{aligned} &\cdot (C + E) \cdot (\bar{C} + \bar{E}) \\ &\cdot (X + \bar{F}) \cdot (X + \bar{E}) \cdot (\bar{X} + F + E) \\ &\cdot (\bar{F} + A) \cdot (\bar{F} + B) \cdot (F + \bar{A} + \bar{B}) \end{aligned}$$

XOR

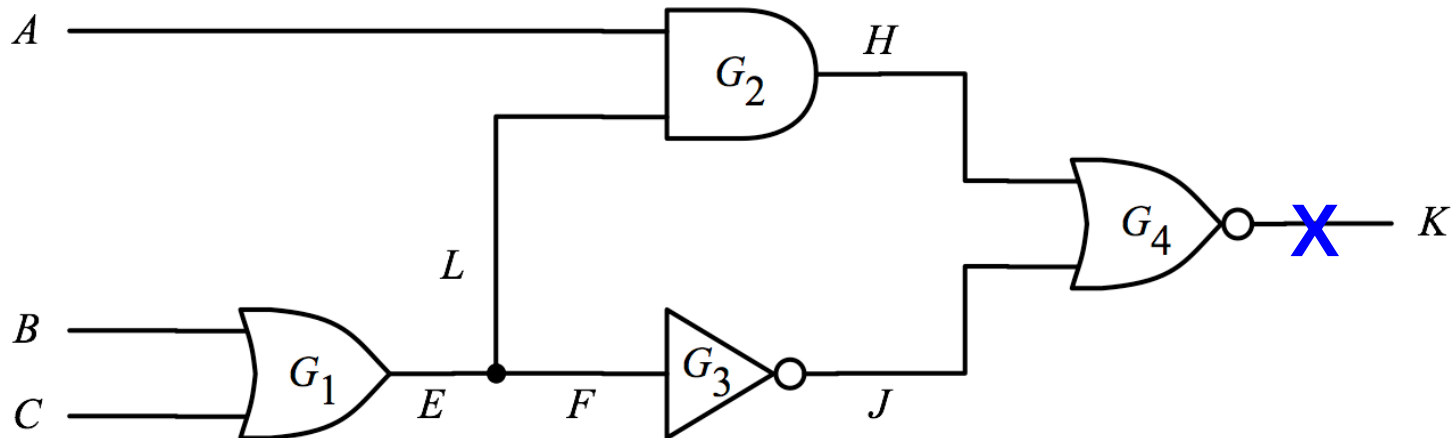
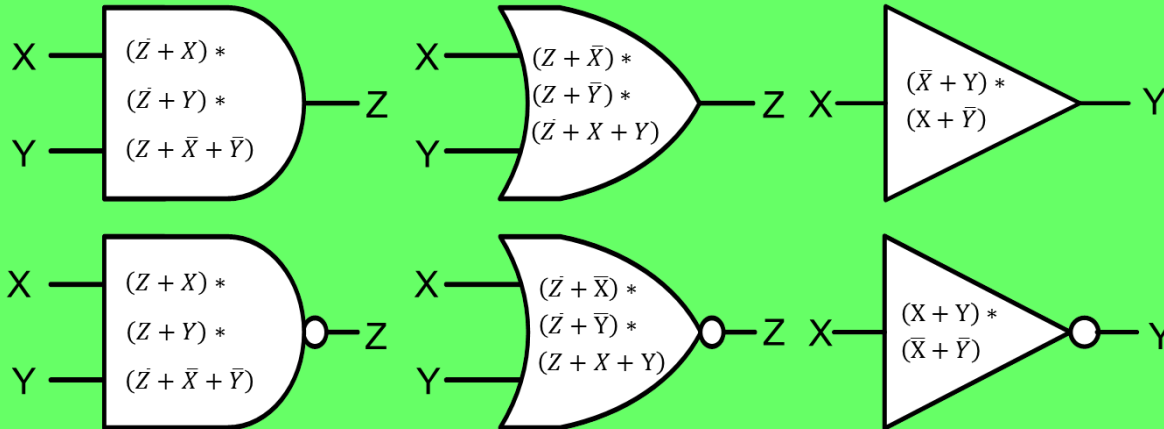
$$\begin{aligned} &\cdot (\bar{X} + x + T) \cdot (X + \bar{x} + T) \\ &\cdot (\bar{X} + \bar{x} + \bar{T}) \cdot (X + x + \bar{T}) \cdot T \end{aligned}$$

=1

Quiz

Q: Please write CNF to generate a test for K stuck-at one fault

A:



Pros and Cons

- SAT-based ATPG
 - + SAT engine is making big progress recently
 - + **proves untestable faults** if CNF is unsatisfiable
 - does not allow **don't cares** in input
 - needs to **redo CNF every time** a target fault is injected
 - does not preserve **circuit structure** information
 - difficult for **multi-valued logic** (such as high impedance)

SAT-based ATPG Not Used in Commercial Tool